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A segmented machine learning modeling approach of social media for predicting occupancy

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Abstract

Purpose – This study aims to predict hotel demand through text analysis by investigating keyword series to increase demand predictions' precision. To do so, this paper presents a framework for modeling hotel demand that incorporates machine learning techniques.

Design/methodology/approach – The empirical forecasting is conducted by introducing a segmented machine learning approach of leveraging hierarchical clustering tied to machine learning and deep learning techniques. These features allow the model to yield more precise estimates. This study evaluates an extensive range of social media–derived words with the most significant probability of gradually establishing an understanding of an optimal outcome. Analyzes were performed on a major hotel chain in an urban market setting within the USA.

Findings – The findings indicate that while traditional methods, being the naïve approach and ARIMA models, struggled with forecasting accuracy, segmented boosting methods (XGBoost) leveraging social media predict hotel occupancy with greater precision for all examined time horizons. Additionally, the segmented learning approach improved the forecasts' stability and robustness while mitigating common overfitting issues within a highly dimensional data set.

Research limitations/implications – Incorporating social media into a segmented learning framework can augment the current generation of forecasting methods' accuracy. Moreover, the segmented learning approach mitigates the negative effects of market shifts (e.g. COVID-19) that can reduce in-production forecasts' life-cycles. The ability to be more robust to market deviations will allow hospitality firms to minimize development time.

Originality/value – The results are expected to generate insights by providing revenue managers with an instrument for predicting demand.

Keywords Deep learning, Hierarchical clustering, Machine learning, Forecasting, Gradient boosting, Social media text

Paper type Research paper

Introduction

The hospitality industry's expansion has coincided with the upsurge in online usage, as the turn of the century, which has resulted in the rapid growth of social media's influence on society, including customer behavior (Gayo-Avello *et al.*, 2013). As social media's influence has grown, industry practitioners have suggested that social media's customer feedback will better position them for ever-changing market conditions. Furthermore, social media has provided access to unstructured qualitative customer data to enhance hotels' understanding of customer needs and improve their yielding ability via demand forecasts. In tandem, this allows hotels to focus more on higher-margin revenue management strategies (Phillips, 2005).

Despite the potential benefits of leveraging social media feedback to improve hotel forecasts, models that incorporated text analysis have received insufficient attention. Furthermore, trifling studies have delved into this area. Pan *et al.* (2012) examined the

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effectiveness of search engine volume data to estimate hotel demand within Charleston, SC. They investigated aggregated search volumes of five keywords incorporated within autoregressive–moving–average (ARMA) models. Although based on a specific destination, the search queries were generic and selected an "optimal" model by comparing the precision of causal econometric models. A similar study conducted by Yang *et al.* (2014) used online traffic volume data of a destination marketing organization (DMO) to predict hotel demand. Yang *et al.* (2014) also used ARMA models, along with their counterparts, ARMAX, while incorporating web traffic data as an explanatory variable.

Likewise, minimal text-based analytical studies exist on hotel demand. This contrasts with abundant diagnostics that forecast tourism demand (Gunter and Önder, 2016; Li *et al.*, 2017; Önder and Gunter, 2016; Yang *et al.*, 2014). These studies have used a range of forecasting methods and distinct leveraged volumes of data. Gunter and Önder (2016) used 10 Google Analytics indicators to predict Vienna's tourist arrivals via a combination of vector autoregressive (VAR) and Bayesian VAR models. The results presented anticipated accuracies when including Google Analytics. Another study conducted by Li *et al.* (2017) discovered that leveraging online search data via introducing a longitudinal search index variable within a dynamic factor model improved forecasts of Beijing tourist volumes. Neither study of Gunter and Önder (2016) nor Li *et al.* (2017) leveraged texts sourced from social media.

This study introduces a modeling framework for incorporating text mining via deep learning techniques on qualitative social media data to improve hotel demand forecasts in light of the aforementioned gaps. Evaluations for the study were sourced from a large-scale hotel resort in a metropolitan city within the USA. Furthermore, this framework for modeling hotel demand developed in multiple stages. First, an extensive range of the hotel's social media keywords was analyzed with the best probability. Besides, using the same text mining approach, keyword ranges were arranged into disparate clusters. Finally, empirical forecasting was conducted by introducing a segmented approach based on hierarchical clustering tied to machine learning techniques. To note, in-depth testing of forecasting variables was not the focus of the study. Instead, this study focused on investigating the modeling framework using common lodging variables.

In so doing, the incremental gains with forecasting performance from each step were demonstrated within the modeling framework as various models were introduced, including the moving average (MA), the autoregressive integrated moving average (ARIMA) and the gradient boosting framework (XGBoost) [1]. Also, quantified social media sentiment metrics will improve lodging forecasts' performance and allow scholars to enhance forecasting accuracy [2], a challenge for demand modeling (Assaf *et al.*, 2019). In addition, by introducing a segmented machine learning approach, scholars and practitioners alike have a framed network for mitigating common data issues when applying learning techniques to big data sets.

This paper is organized as follows. Section 2 discusses the relevant literature and, based on findings and gaps therein, Section 3 describes the study's proposed methodology in detail. Section 4 presents the empirical study and explains and discusses the evaluation results. Finally, Section 5 presents the conclusions and outlines limitations and future research.

Literature review

Forecasting with online data

In hospitality and tourism literature, most studies use historical patterns of time series to forecast demand. The most popular ones include the vector autoregressive models (VAR), vector error correction model (VECM) or Bayesian technique (BVAR), ARIMA, analytical

hierarchy process (AHP), exponential smoothing (ETS), time-varying parameter models (TVP), trigonometric regressors to model multiple-seasonalities with Box-Cox transformations, ARMA errors, trend and seasonality (TBATS) and choice modeling (Witt and Witt, 1992; Crouch, 1994; Schwartz and Cohen, 2004; Li *et al.*, 2005; Wong *et al.*, 2006; Yüksel, 2007; Zakhary *et al.*, 2011; Haensel and Koole, 2011; Song *et al.*, 2013; Gunter and Önder, 2016; Pereira, 2016; Ampountolas, 2019). Researchers also have used hybrid forecasting models by combining econometrics and data mining techniques (Pai *et al.*, 2014; Sun *et al.*, 2016).

Social media's advent offers researchers an enormous data source to incorporate into such models while improving forecasting. However, introducing an unstructured volume of data brings new challenges. If it is to increase the precision of forecasts, researchers need to consider overfitting or generalization issues that may transpire. Multiple studies have examined the incorporation of internet search volume data into forecasting models outside of hospitality and tourism industries and investigated its influence on consumption levels for future purchases (Carrière-Swallow and Labbé, 2013; Vosen and Schmidt, 2011; Choi and Varian, 2012), unemployment rate (Askitas and Zimmermann, 2009) and stock market index (Preis *et al.*, 2013) as a forecasting variable.

Big data and demand forecasting models have become relevant within hospitality and tourism research in recent years. Choi and Varian (2009) conducted some of the earliest studies to use external search data for forecasting customer demand (2009; 2012)[a,b]. They used Google Trends data from the Hong Kong Tourism Board to estimate customer arrivals from key markets including, the USA, Canada, Great Britain, Germany, France, Italy, Australia, Japan and India. They concluded that basic seasonal AR models and fixed-effects models outperform traditional models without external predictors. Pan *et al.* (2012) furthered the research by examining more advanced forecasting methods, ARMA and ARMAX to forecast hotel room demand. They also examined causal econometric models such as the autoregressive distributed lag (ADL), TVP and VAR. The result was the ARMAX models outperformed the ARMA models, concluding that the value of search volume data favors accurate prediction for forecasting demand. Likewise, when compared against the causal econometric models, the ARMAX models generated more precise estimates.

Gunter and Önder (2016) used Google Trends and incorporated 10 Google Analytics indicators obtained from the Viennese DMO to predict Vienna's tourist arrivals. To overcome the over-parameterization big data might cause, their study used a combination of VAR and Bayesian VAR models, which resulted in a factor-augmented vector autoregressive (FAVAR) model. Furthermore, they benchmarked the advanced models with standard forecasting models, including MA or ETS. The results confirmed that standard forecasting models were more accurate for shorter horizons. On the other hand, the combined advanced models generated more precise results for longer horizons when including predictive information from Google Analytics.

Another study conducted by Pan and Yang (2017) applied time-series models to forecast weekly hotel occupancy in Charleston, SC, using popularity data from sources such as search engine queries, website traffic and weather. Their study was based on ARMAX models, an extension of the ARMA model's family and they incorporated big data variables as direct predictors. The ARMAX models were more accurate when including the external predictors. A similar study by Park *et al.* (2017) examined tourists' arrivals from Japan to South Korea by using combinations of select keywords from Google search terms. The scholars tested a standard ARIMA model, a seasonal ARIMA model with exogenous variables and Holt Winter's seasonal smoothing model. They concluded that models

incorporating Google search terms were better than standard time-series models with shorter-term horizons.

Li *et al.* (2017) also implemented search engine volume data, including tourist numbers and hotel occupancy, to forecast demand in Beijing, noting that highly correlated data in the forecasting model create challenges and reduce generalization. To counter this, the authors proposed a framework to design a composite search index, which they tested within the generalized dynamic factor model (GDFM) and principal component analysis (PCA) based indices. The results indicated that the introduced approach enhances accuracy better than the benchmark models such as a standard time series model or one with an index created by principal component analysis.

To conclude, in a recent study, Gričar and Bojnec (2019) introduced a cointegrated (C) VAR model based on aggregated time-series prices of short-stay accommodations in Slovenia. The overnight stays for international tourists formed a positive relationship with short-stay accommodation prices.

Forecasting in an advanced booking environment

The hurdles for revenue management forecasting are distinctive, considering the dynamic booking windows. Because the shift in reservation leads might fluctuate, Tse and Poon (2015) used a hotel reservation data set to examine a booking curve's trends in forecasting. Their findings indicated that patterns identified in the previous data could be worthwhile in forecasting accuracy. Likewise, in a current study, Webb *et al.* (2020) have introduced a multi-layer perceptron (MLP) neural network model to address the challenges of booking window shifts. These shifts present concerns while developing forecasting models. The authors have used reservation accumulations over a period as an input vector. The results show that the MLP achieves high accuracy in the examined scenarios, indicating that it may contribute to a viable forecasting technique in the advanced booking environment.

Corresponding to the previous analysis, in their work, Fiori and Foroni (2020) tested a stochastic framework by using data from a hotel in Italy to construct prediction intervals for a reservation-based pickup forecasting approach. They extended the multiplicative pickup technique with two variants, namely, distribution-free and the Generalized Linear Models. The results indicated better prediction intervals comparable to classical time series methods. Hence, it confirmed that managerial decisions could not depend on point predictions of future room demand. Finally, earlier in another work, Schwartz *et al.* (2016) proposed a model using the hotel competitive set's predicted occupancy as an input of the individual property forecast using a recursive approach to improve the hotel's daily occupancy forecast accuracy. The results suggested that the forecasting accuracy can be enhanced by exercising either a genetic algorithm or the simple linear regression model.

Although these surveys incorporated search volume data within econometric models, one distinct characteristic is our analysis of extensive and inconsistent data set with machine learning models. We attempt to include source volume figures and data from social media to optimize daily demand forecasting accuracy.

Sentiment analysis

Sentiment analysis studies people's opinions toward the services or brands or emotions and captures a person's opinion, whether a positive, neutral or negative sentiment (Liu, 2015). It has been treated as a derivative of Natural Language Processing (NLP) because it focuses on written text (Hu and Liu, 2004). As existing research in the field has grown, it has been carried out at three levels of granularity (Liu, 2015): document-level classification expressed on positive or negative opinions (Turney, 2002; Pang and Lee, 2004), sentence-level

classification that denotes a positive, neutral or negative opinion (Hu and Liu, 2004) and, more recently, the aspect level inherent by people's liked or disliked attributes (Agarwal *et al.*, 2009; Liu, 2010). Firms leverage machine learning algorithms on mined text to better understand their customers' behaviors and shifting trends.

There are two key methods for evaluating sentiment in text: corpus-based and dictionary-based approaches (Hu *et al.*, 2013). The former uses a clustering algorithm that segments based on the patterns of explicit and implicit expressions, along with the frequency of co-occurring patterns in the text (Su and Markert, 2008; Chiu *et al.*, 2016; Liu, 2015). The latter approaches include pre-defined lexicons to improve sentiment coverage (Liu, 2015). A benefit of dictionary-based approaches is that they include word categories to complement the corpus-based approach, as dictionaries define adjectives using attributes (Fei *et al.*, 2012). Lexicons are also not restricted by a particular corpus; there are also sentiment lexicons built on an array of positive and negative word seeds in WordNet (Hu and Liu, 2004). Sentiment analysis uses the collection of words to predict sentiment even beyond sentence levels such as document, paragraph or attribute levels. Hence, this method has robust applications in statistical models.

Forecasting models

Different econometric models have been leveraged within the literature on forecasting hospitality demand. The fundamental question among scholars has been how to advance accuracy with proposed frameworks while still producing a well-generalized model (i.e. minimal overfitting). Therefore, innovative models ensure higher accuracy and capitalize on their effectiveness. Motivated by the preceding literature, we examine multiple approaches while presenting a modeling hotel demand framework. Furthermore, we adopt a combination of basic forecasting models and machine learning techniques. We start by introducing a naïve model, followed by a traditional ARIMA model, before encoding an advanced machine learning model, the XGBoost, to predict hotel occupancy.

Traditional models

The moving average (MA) is an application of a stochastic process. We consider the weighted average of past observations in a time series distribution while forecasting the value based on the average of the random variable (Hyndman and Athanasopoulos, 2018):

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \ldots + \theta_q \varepsilon_{t-q}, \tag{1}$$

where ε_t is white noise in an MA(q) model of order q.

The ARIMA model represents an important class of ARMA linear models often used in tourism forecasting. ARIMA models describe stationary, non-stationary and seasonal order processes (p, d, q) (Hyndman and Athanasopoulos, 2018). A common obstacle in applying ARIMA models is that the order selection process is considered subjective and tough to apply (Hyndman and Athanasopoulos, 2018). For this reason, several attempts have been made to automate and simplify the ARIMA modeling process; this is even more difficult in industries that exhibit strong seasonality with significant demand volatility therein such as hospitality. The forecasting procedures include the ARIMA process as:

$$y_t = \sum_{\theta=1}^{p} \alpha_i y_{t-i} + \sum_{j=1}^{q} \beta_j \varepsilon_{t-j} + \varepsilon_p, \tag{2}$$

where ε_t represents the uncorrelated random noise of each time series, α_i (θ = 1, 2, . . ., p) and β_j (j = 1, 2, . . ., q) is the $m \times m$ coefficient matrices of the multivariate ARIMA model (Hyndman and Athanasopoulos, 2018).

Machine learning models. Machine learning-based systems are becoming more popular in research applications in most fields. One such learning algorithm that has gained traction is XGBoost, introduced by Chen and Guestrin (2016). It is built on a gradient algorithm (Friedman *et al.*, 2001), which provides scalable tree boosting and operates parallel processing. The boosted trees are constructed to address regression and classification trees. Notably, they are designed to be flexible and efficient while optimizing the outcome's predictive value (Hastie *et al.*, 2009). Gradient boosting also prepares the boosted trees by obtaining the feature scores and displaying their importance to the training model when applied to historical data.

To improve model generalization with XGBoost, Chen and Guestrin (2016) introduced an adjustment to the cost function, making it a "regularized boosting" technique:

$$L(f) = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{m=1}^{M} \Omega(\delta_m) \tag{3}$$

with

$$\Omega(\delta) = \alpha|\delta| + \frac{1}{2}\beta \parallel w\|^2, \tag{4}$$

where $|\delta|$ refers to the number of leaves of the classification tree δ and w is the values attributed to each leaf. The regularizer Ω penalizes the model's complexity and can be interpreted as the ridge regularization of coefficient β and Lasso regularization of coefficient α. Hence, the cost function falls back to the traditional gradient tree boosting when the parameter is set to zero (Omar, 2018).

Gradient boosting constructs the boosted trees to obtain the feature scores. The algorithm counts the importance of "frequency," "cover" and "gain" in each feature, starting from a single leaf and adding branches to the tree. Frequency represents the relative times a feature occurs; higher frequency scores indicate the feature is used more often in the boosting process. The cover presents the number of observations related to the feature. Gain refers to the initial reference factor of contribution to every constructed tree. Each leaf consists of adding first- and second-order derivatives to evaluate the standardized objective and maximize its decline when searching for splits (Friedman *et al.*, 2000):

$$L(f) \approx \sum_{i=1}^{n} \left[(L(y_i, \hat{y}_i)) + g_i \delta_t(x_i) + \frac{1}{2} h_i \delta_t^2(x_i) \right] + \Omega(\delta_t) \tag{5}$$

where $g_i = \partial_{\hat{y}} L(\hat{y}_i, y_i)$ and $h_i = \partial_y{}^2 L(\hat{y}_i, y_i)$. By removing the constant term, we receive the following approximation of the objective at step t:

$$\hat{L}(f) = \sum_{i=1}^{n} \left[g_i \delta_t(x_i) + \frac{1}{2} h_i \delta_t^2(x_i) \right] + \Omega(\delta_t), \tag{6}$$

By defining I_j as the instance set at leaf j and expanding Ω, we can rewrite the equation as:

$$\hat{L}(f) = \sum_{j=1}^{T} \left[\left(\sum i \varepsilon I_j g_i \right) w_j + \frac{1}{2} \left(\sum i \varepsilon I_j h_i + \beta \right) w_j^2 \right] + \alpha |\delta| \tag{7}$$

XGBoost does not use entropy or information gain for splits in a decision tree but instead uses the following gain:

$$G_j = \sum i \, \epsilon I_j g_l, \tag{8}$$

$$H_j = \sum i \, \epsilon I_j h_l, \tag{9}$$

$$\text{Gain} \quad = \frac{1}{2}\left[\frac{G_L^2}{H_L + \beta} + \frac{G_R^2}{H_R + \beta} - \frac{(G_R + G_L)^2}{H_R + H_L + \beta}\right] - a, \tag{10}$$

where the first term is the left child's score, the second is the score of the right child, if we do not split the third the score; α is the complexity cost if we add a new split (Friedman *et al.*, 2000; Omar, 2018). In this study, by implementing XGBoost, we sort the attributes, revise all possible values of the split and compute the gain.

Hierarchical clustering. Machine learning models often include many parameters, leading to generalized models that overfit the training data. However, sufficient parameters are critical to capture nuances and prevent under-fitting issues. With the use of regularization, poor generalization can often be avoided (Voulgaris, 2016). The regularization of models uses parameters differently by merging them while simultaneously fine-tuning them. The resulting models are more generalized while also producing more specific calculations.

To overcome the above challenges, in addition to using regularization, we also leverage a hierarchical clustering method to reduce the data before the machine learning training is processed. The hierarchical method is sub-par to operate with a large data set. At the same time, the reducing and clustering using the (BIRCH) hierarchical method introduced by Zhang *et al.* (1997) is ideal for examining large data sets of continuous variables (Halkidi *et al.*, 2001). Moreover, hierarchical clustering algorithms handle noise and outliers to determine the optimal number of clusters (Horng *et al.*, 2011). Therefore, we applied hierarchical clustering to a dense data set and produced a high-quality, reduced data set to represent the prototype.

Data and methodology

The study's data is sourced from the hotel property management system (PMS), dated between January 2016 and September 2019, of a destination resort in a major US market. The resort's PMS contains booking information associated with the room and customer segments stored at room level, including, but not limited to, the distribution channel, VIP status and room type. The hotel's PMS system had 102,000 observations or booked rooms from January 2016 through September 2019. In addition to the PMS data, we collected over 3,500 tweets from Twitter and SocialMention that referenced the resort over the same period. Twitter is a data source that enables users to post and communicate their opinion in real-time on social media (Bughin, 2015; Rao and Srivastava, 2013).

We set January 1st, 2016, through June 30th, 2019, to develop the models. Subsequently, the remaining timeframe from July 12th, 2019, to September 30th, 2019 became an out of time (OOT) sample to test each of our models' fit. For any that used a validation data set (e.g. machine learning) within their development, the validation data set was split from the training data set. Using common revenue management metrics, we also assessed each model's forecasts based on their performance with 4 forecast horizons on rooms occupied of 1 day, 4 days, 7 days and 14 days, all applied to the OOT sample. That allowed us to assess the stability and strength of each model's forecasts. Moreover, we used one-sided Diebold-Mariano (DM) tests to measure incremental predictive accuracy between each sequential model (Diebold and Mariano, 1995). The traditional revenue management metrics to assess

performance were, mean absolute deviation (MAD), the mean square error (MSE), the root mean square error (RMSE) and the mean absolute percentage error (MAPE).

$$\text{Mean absolute deviation}: \quad \text{MAD} = \text{mean}(\,|e_t|\,), \tag{11}$$

$$\text{Mean Square Error}: \quad \text{MSE} = \text{mean}\left(e_t^2\right),, \tag{12}$$

$$\text{Root mean squared error}: \quad \text{RMSE} = \sqrt{\text{mean}\left(e_t^2\right)}, \tag{13}$$

$$\text{Mean absolute percentage error}: \text{MAPE} \quad = \text{mean}(\,|p_t|\,), \tag{14}$$

where e_t represents the forecast errors and p_t the percentage errors.

The first model uses the naïve approach by applying last year's same-day occupancy to the future forecast. While this is not optimal, it is popular among revenue management practitioners (Hyndman and Athanasopoulos, 2018). For this reason, we set the naïve approach as the preliminary.

The second model is the ARIMA time series. As that requires an examination of the data's trending behavior to explore the stationarity (Montgomery *et al.*, 2015), we started with Dickey-Fuller (ADF) and Phillips-Perron (PP), both coefficient tests for detecting non-seasonal unit-roots (Dickey and Fuller, 1979; Phillips and Perron, 1988), to determine the data stationarity. Besides, we have used the monthly Hylleberg–Engel–Granger–Yoo (HEGY) test (Hylleberg *et al.*, 1990), a unit root test for seasonal frequencies without anticipating unit-roots were indicated at all seasonal frequencies (Beaulieu and Miron, 1993). The null hypothesis for the scheduled data HEGY test is for a unit root at cycles for frequencies, the seasonal unit roots at frequencies π, $\pm\pi/2$, $\mp 2\pi/3$, $\pm\pi/3$, $\mp 5\pi/6$, $\pm\pi/6$ are present. Therefore, following the process developed by Beaulieu and Miron (1993), seasonal unit roots for monthly data are. To then determine the amount of auto-regressive and moving average factors, we use a combination of the diagnostic plots of the autocorrelation function (ACF) and partial autocorrelations functions (PACF) while examining the statistics of Akaike Information Criterion (AIC) and Schwarz Information Criterion (SIC) from multiple iterative ARIMA models. Ultimately, the model that has the best combination of fit scores was selected.

With our initial two base models set, we then built our first iteration leveraging XGBoost. For all duplications of XGBoost models, we split a stratified random sample of 25% from the training data set into a validation data set to confirm the parameters' tuning. We subsequently set daily occupied rooms as our dependent variable and included a combination of auto-regressive lags and moving average variables of occupied rooms, seasonality attributes and daily pickups into the algorithm. In total, about 50 independent inputs were included. Of note, social media data was not included for the first iteration of the XGBoost model.

Also, as the learning algorithm is applied at the macro level as daily occupied rooms, we limited the trees' depths and cross-validations to minimize overfitting issues (Lusa, 2017; Park and Ho, 2019; Reyzin and Schapire, 2006). Next, we assess variable importance in the learning algorithm via matrix importance plots. To mitigate rare events bias where input can contribute to poorly generalized models (Lusa, 2017; Ghojogh and Crowley, 2019; Li *et al.*, 2019), we limit the data's dimensionality by removing variables with less than 0.50

gains. We investigate the statistics of MAPE, RSME, MSE and MAD between the training data set and the validation data set to assess the overall model fit and generalization (Ghojogh and Crowley, 2019).

Next, we conducted a sentiment analysis on the longitudinal social media data. For this study, we leveraged three general-purpose dictionaries: the AFINN (Nielsen, 2011), Bing (Liu, 2010) and the NRC (Mohammad and Turney, 2013). The dictionaries' combination assigned emotional scaling to the posts' words and calculated sentiment based on relative trajectories. Overall, the three lexicons provided a base of more than 20,000 words. In this study, there was a total of 3,500 tweets, from which, after cleansing, we reviewed 2,600. After assigning a sentiment score, we summarized the daily sentiment scores versus the number of tweets. The sentiment-based variables are then added as an independent input to the XGBoost model. The learning algorithm is then returned with the added input, and model fit statistics from the updated XGBoost model are compared to the first statistics without the variables.

The rooms in the PMS data were segmented into homogenous clusters within the initial learning algorithms. To derive the clusters for the segmented XGBoost model, we leverage hierarchical clustering in the resort's PMS data. As the PMS data contained over 100,000 observations and hierarchical clustering methods struggle to process larger data sets (Gilpin *et al.*, 2013), we used a process to solve this while retaining the necessary booking information. We started with summarizing booking curves, seasonal patterns and daily distributions of booked rooms for each custom attribute. In total, the PMS data had 24 customer attributes. Next, we standardized each attribute's metrics and introduced them into the median algorithm's hierarchical clustering. To assess the ideal number of clusters, we used a combination of the explanatory statistics of semi-partial R^2, the median distance between clusters and the cubic clustering criterion.

With the clusters set, we returned the parameter inputs from the XGBoost model before summarizing each segment-based forecast into a final segmented XGBoost forecast. Finally, we then compared the results with the previous XGBoost model iteration without segments.

Results

In examining the longitudinal data, we discovered seasonality trends with the peak season being Spring at 84% occupancy and Fall as the off-peak season with 73%. March had the highest occupancy at 90%, while September had the lowest; over the 3 years, it averaged 65%. Also, the resort was in more demand over the weekends, with an 82% occupancy, compared to weekdays with 75%. On Monday, the business week's start coincided with the lowest occupancy at 72% compared to Saturdays, which had the highest at 87%.

Naive and autoregressive integrated moving average forecasts

We start by applying the naïve approach to the OOT sample. Over the 92 days, the naïve approach had a 15.4% MAPE, as well as an MSE of 11.9 and an RMSE of 3.4 points. Results of higher MAPE and MSE are unexpected, as sudden seasonal shifts – or even slight shifts from year to year – can reduce the precision of naïve forecasts.

We concluded the data is not stationary by examining unit-roots with the Dickey-Full (ADF) test and Phillips-Perron tests (PP) for stationarity but corrected the issue by introducing year over year (YoY) differencing into the ARIMA model. Table 1 indicates that in the monthly data HEGY test, the null hypothesis of a unit root's presence at frequency 0 cannot be rejected as the data fail to discard unit-roots for at least one seasonal frequency. Likewise, we investigated the ACF plot to further confirm stationarity with no decay of the ACF plot's autoregressive lags. Through the plots of the ACF and PACF and by running six

Table 1. ARIMA model selection

	Chosen model					
Model coefficients	*(1)*	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*
MU	0.65	0.65	0.64	0.67	0.64	0.67
MA(1)	−0.07	−0.07	−0.08	−0.09		
MA(2)	0.09	0.09**	0.07	0.07**	0.12**	0.15**
MA(7)		0.10				
MA(8)	−0.06	−0.06	−0.06	0.04		
MA(14)	0.04	0.04			0.04	
MA(28)	−0.66**	−0.66**	−0.66**		−0.66**	
AR(1)	0.53**	0.53**	0.51**	0.50**	0.59**	0.58**
AR(2)	0.000		−0.004	−0.002		0.03
AR(7)		0.10	0.001	0.12		0.07
AR(28)	−0.57**	−0.57**	−0.56**		−0.56**	
Fit statistics						
AIC	7,190	7,193	7,192	7,196	7,187	7,193
SIC	7,233	7,246	7,241	7,230	7,216	7,217

HEGY test results		*Frequencies*						
	Lag Order	0	π	$\pm\pi/2$	$\pm 2\pi/3$	$\pm\pi/3$	$\pm 5\pi/6$	$\pm\pi/6$
Rooms occupied	1	0.95	−2.39	0.58*	2.26*	2.15*	3.31*	2.70*

Notes: **Indicates significance at $\alpha = 0.05$ level, *indicates significance at $\alpha = 0.10$ level, AIC is akaike information criterion, SIC is the Schwarz information criterion, the HEGY test includes an intercept term, a time trend and seasonal dummy variables, the monthly HEGY test's critical values have been presented according to the structure provided by Beaulieu and Miron (1993). The optimal lag order for HEGY is determined by BIC

ARIMA models, we determined 3 moving average coefficients (q = 2, 14 and 28) and 2 autoregressive coefficients (p = 1 and 28) that exhibits the lowest AIC and SIC scores of 7,187 and 7,216 as shown in Table 1.

Forecasting with the chosen ARIMA model applied to the OOT sample resulted in a 48% improvement in the MAPE (10.4% vs 15.4%) when used on a 1-day horizon over the naïve approach and a 5% improvement in the MAPE when used on a 14-day horizon (14.8% vs 15.4%). According to Diebold-Mariano (DM) tests, the chosen ARIMA model forecasts occupancy at an $\alpha = 0.05$ level with the shorter horizons of 1 day and 4 days. However, this is not the case for horizons of 7 days and 14 days, as DM tests show the ARIMA model forecast of occupied rooms is more accurate at an $\alpha = 0.10$ level for 7 days and dismal accuracy for a 14-day horizon. The smaller MAPE improvement seems to be influenced by larger potential errors, as the ARIMA had a larger MSE of 12.2 points, compared to the naïve approach's MSE of 11.9 points.

Gradient boosting framework w/out social media

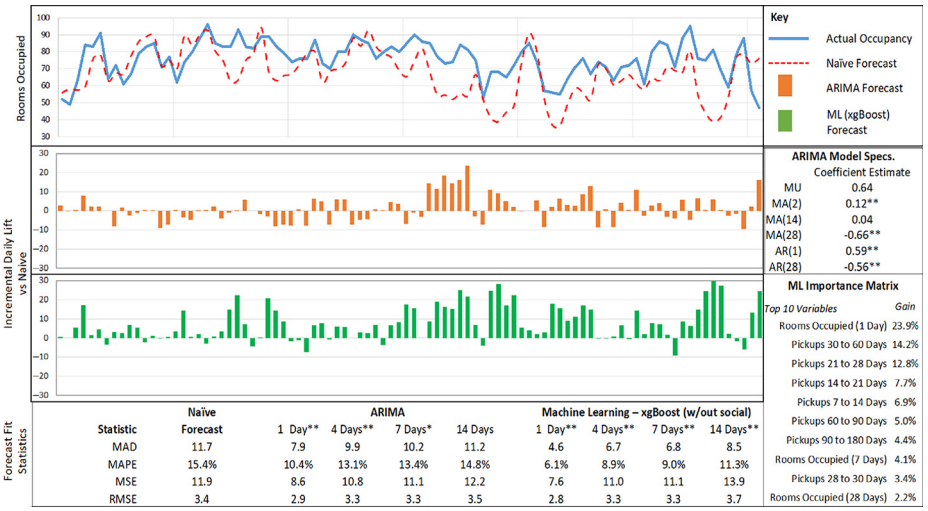
After applying an ARIMA forecast, we input 50 variables into an XGBoost learning algorithm to predict daily occupied rooms. We further split 25% of the training data set into a validation data set. By finetuning multiple iterations and investigating gains reported in the ML Importance Matrix, we finalized half (25) of the independent variables. We mitigated potential overfitting issues by reducing dimensionality in the data, as the gap in the RSME between the training and validation data set was minimized (Ghojogh and Crowley, 2019). Between the training and validation, the gap in RMSE data sets for the learning algorithm

with 50 variables is 1.2 points (1.8 points vs 3.0 points), compared to a gap in the RMSE of only 0.6 points (2.1 points vs 2.7 points) for the 25 remaining variables.

Of the remaining variables in the model, most (7 out of 10) of the contributing variables are derived from daily pickups. The one with the most significant contribution in gains was a 1-day lag of rooms occupied, with a gain of almost a quarter (23.9%). The next most considerable lag-based variable is a 7-day lag of rooms occupied, with only a 4.1% contribution. Forecasting with the XGBoost model applied to the OOT sample resulted in a 70% improvement in the MAPE (6.1% vs 10.4%) when used on a 1-day horizon over the ARIMA model and a 30% improvement in the MAPE when used on a 14-day horizon (11.3% vs 14.8%). Compared to the naïve forecast, the DM test statistics show that the XGBoost model is more accurate at predicting occupancy at an $\alpha = 0.05$ level for all tested time horizons of 1, 4, 7 and 14 days. The results and fit statistics of the other XGBoost forecast compared to the ARIMA and the naïve forecast are shown in more detail in Figure 1.

Sentiment analysis

Next, we applied three general-purpose sentiment dictionaries – the AFINN, Bing and NRC – to the 2,600 cleansed tweets issued between January 2016 and September 2019. The resort received more positive sentiments at approximately a 3 to 1 ratio, with 2,350 positive tweets compared to 750 negative tweets [3]. Of the most tweeted lodging-based terms, we find *spa* ($n = 145$), *view* ($n = 135$) and *rooms* ($n = 133$) to be most associated with positive sentiments, while *club* ($n = 84$), *rooms* ($n = 73$) and *service* ($n = 63$) are most associated with negative ones, as shown in Figure 2. The average absolute positive sentiment scores reported among the top 10 associated lodging terms are lower (average score = 0.22) than the negative sentiment (average = −0.25). These lower scores indicate that negative sentiments tend to be associated with stronger emotions. In other words, if a customer has an issue with their room, they either expressed more negative verbiage or used more definite negative connotations in their complaints on social media.



Statistic	Naïve Forecast	ARIMA 1 Day**	ARIMA 4 Days**	ARIMA 7 Days*	ARIMA 14 Days	Machine Learning – xgBoost (w/out social) 1 Day**	4 Days**	7 Days**	14 Days**
MAD	11.7	7.9	9.9	10.2	11.2	4.6	6.7	6.8	8.5
MAPE	15.4%	10.4%	13.1%	13.4%	14.8%	6.1%	8.9%	9.0%	11.3%
MSE	11.9	8.6	10.8	11.1	12.2	7.6	11.0	11.1	13.9
RMSE	3.4	2.9	3.3	3.3	3.5	2.8	3.3	3.3	3.7

Figure 1. Naïve vs ARIMA vs XGBoost w/out social media

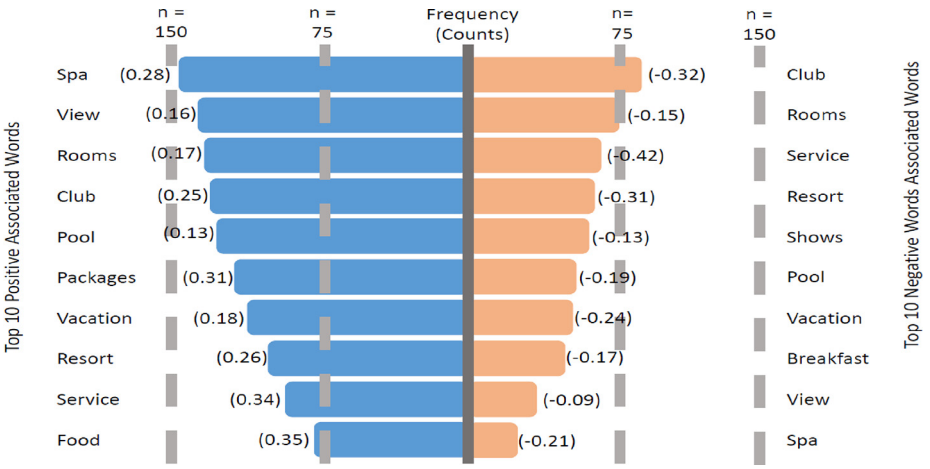


Figure 2.
Sentiment results

Gradient boosting framework w/social media

We summarized the sentiment results into moving average scores and add them as independent input into the reduced 25-variable XGBoost model. The XGBoost learning algorithm is then returned to keep only independent variables above 0.5 gains. The resulting XGBoost model contains two in the top 10 of contributing variables to the XGBoost model, being the average sentiment score from 14 to 28 days prior ranked at #8 with 3.6% total gains and the one from 7 to 14 days prior ranked at #10 at 3.2%. In total, sentiment variables contribute 8.2% to the model according to the Importance Matrix. The other top variables follow the prior iteration of the XGBoost model without social media.

Applying the returned XGBoost model with social media inputs to the OOT sample results in considerable improvements to MAPE within each forecast horizon. They also provide marginal improvements to RMSE, as compared to the XGBoost model without social media inputs. Specifically, the retuned model shows an 11% improvement in the MAPE (5.4% to 6.1%) and a 1% advance in RMSE (2.7 to 2.8) when used on a 1-day horizon. Even though the XGBoost with social media inputs showed an 11% improvement with MAPE on a 1-day horizon over XGBoost without social media input, the DM test proved to be insignificant at an $\alpha = 0.05$ level. However, Social Media inputs helped with longer time horizons as the DM tests showed that the XGBoost with social media inputs predicted rooms occupied at an $\alpha = 0.05$ level more accurately than XGBoost without it for 4, 7 and 14 days. The considerable development in MAPE and lack of improvements to RMSE indicate the XGBoost model with social media inputs has a larger variation in its ability to forecast daily rooms occupied than the XGBoost model without it. Even though the XGBoost model with social media input has more significant swings, it still moves the forecast closer to the actual rooms occupied more often than the XGBoost model. Additional details on forecasting results for the XGBoost model w/social input are detailed in Figure 3.

The more substantial variances in the residuals with the forecasted observations in the XGBoost model are might be due to noise from less critical variables such as the sentiment trends that could be more correlated with daily rooms occupied during the learning period than during the forecasted time span. The loss in correlation across periods can also be explained by a shift in the distribution of customer segments that are more influenced by social media trends. For example, a distribution of the segments reveals that online travel

Forecast Fit Statistics

Key	ML (xgBoost) w/Social Forecast
**	DM-Test significant at α = 0.05 level
*	DM-Test significant at α = 0.10 level

Machine Learning - xgBoost (w/Social)

Statistic	1 Day*	4 Days**	7 Days**	14 Days**
MAD	4.1	5.2	5.4	7.1
MAPE	5.4%	7.3%	7.4%	9.5%
MSE	7.4	10.5	11.3	13.2
RMSE	2.7	3.2	3.4	3.6

ML Importance Matrix

Top 10 Variables	Gain
Rooms Occupied (1 Day)	22.2%
Pickups 30 to 60 Days	13.0%
Pickups 21 to 28 Days	12.4%
Pickups 14 to 21 Days	7.6%
Pickups 7 to 14 Days	6.2%
Pickups 60 to 90 Days	4.7%
Pickups 90 to 180 Days	4.4%
Sentiment Score 14 to 28 Days	*3.6%*
Pickups 28 to 30 Days	3.3%
Sentiment Score 7 to 14 Days	*3.2%*
Total Gains from Social Derived Variables	*8.2%*

Notes: *Social media derived parameters are noted in italics, incremental lift and Diebold-Mariano (DM) test statistics calculated by taking the mean absolute deviation (MAD) gains in the ML xgBoost forecast with social data vs. ML xgBoost forecast w/out social data.*

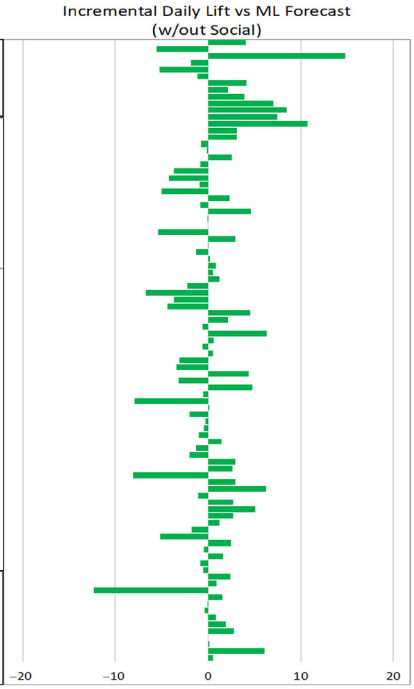


Figure 3. Non-segmented XGBoost w/social media

agency (OTA) bookings grew from 38% in the training period to below 50% in the OOT period. Likewise, direct site bookings decreased from 42% to 28% for the same respective time. If customers booking through the site react differently to social media postings than OTA customers, the shift will introduce noise when the XGBoost model is applied to the forecasted period, leading to more significant residuals.

Hierarchical clustering of bookings

To mitigate the introduction of noise in the XGBoost model, the data has unique customer segments based on their booking patterns and the timing of their stays. Six summarized variables and 28 customer attributes are placed in a hierarchical clustering algorithm using the median differencing approach. From the hierarchical results, 4 clusters are the optimal number of segments, as this is where the cubic clustering criterion metric peaked and where both the semi-partial R^2 and median distancing between clusters showed minimal improvement. These are split across distribution channels as each holds unique booking patterns and seasonality attributes. The segments are OTA, global distribution systems (GDS) and phone channels; direct web bookings; walk-ins; and groups. Figure 4. represents the clusters in a dendrogram, which records the classifications of merges or splits in the various hierarchical clustering steps.

Segmented gradient boosting framework w/social media

Next, we returned the XGBoost model with social media inputs on all 4. The differences between each segment yield noteworthy results that reaffirm the forecasting approach, as shown in Figure 5. Even though each segment's best contributor is a 1-day log of rooms

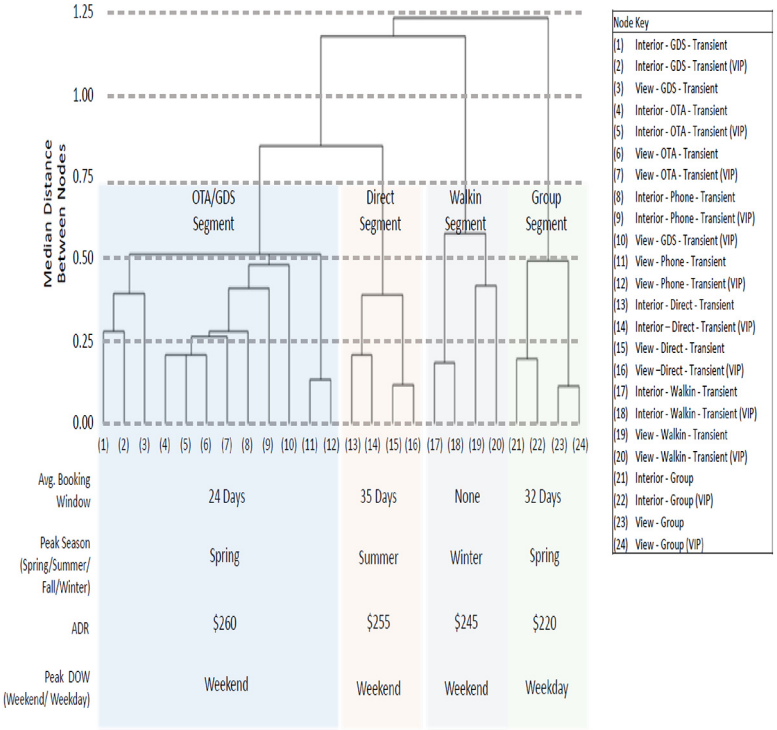


Figure 4.
Hierarchical clusters (dendrogram)

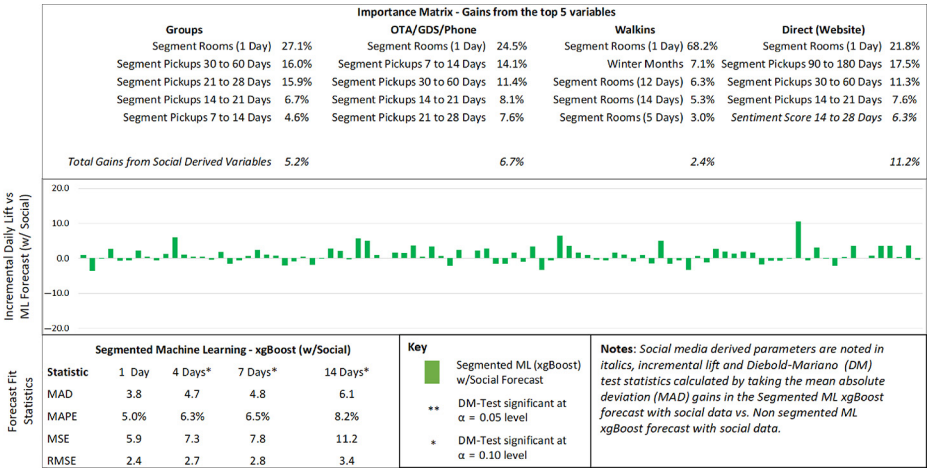


Figure 5.
Segmented XGBoost w/social media

occupied, its contributions differ. The previous day's room occupation contributes almost three quarters (or 68.2% gain) to the walk-in segment while contributing two-thirds less (or 21.8% gains) to the direct website customer. More recent daily pickups show a greater influence in both group segments and the OTA/GDS/phone segment, with pickups within

the previous 28 days contributing 29% and 31% to total gains. In contrast, historical daily pickups contribute more to the direct web booking segment, between 90 and 180 days prior, comprising 17.5% of total gains. The total contribution trend also differs by the customer, with social media sentiments being most predictive within the direct web booking, at 11.2% of the total gains and least predictive within the walk-in's segment, at only 2.4%.

The ability to vary the input for each customer segment mitigates noise issues caused by insignificant variables in a considerable improvement to the forecasted 1-day horizon of RMSE (12%), compared to the non-segmented XGBoost forecast with social media input (2.4% vs 2.7%). The forecasted MAPE improvement is minimal when the segmented XGBoost is compared to the non-segmented XGBoost – only by 8% (5.0% vs 5.4%) when used on a 1-day horizon. The DM tests further confirm the improvement to be irrelevant at an $\alpha = 0.05$ level for the 1-day horizon. For longer time horizons of 4, 7 and 14 days, the DM tests showed the segmented XGBoost predicts occupancy at an $\alpha = 0.10$1. These results show the segmented XGBoost model corrects the large misses or residuals in the forecast but offers limited precision.

Summarized results

To summarize, each sequential model introduced here added incremental improvements for at least one of the four time-horizons tested, as shown in the Diebold-Mariano (DM) in Table 2. The DM tests show that all machine learning algorithms improved the forecasts at an $\alpha = 0.05$ level over the ARIMA and the Naïve forecasts for all time horizons. Besides, the DM tests prove that social media entered into an XGBoost model may not improve the accuracy of shorter time horizons than XGBoost without it. To this extent, the combination of segmenting the customer base and adding social media input adds significant incremental predictive accuracy at an $\alpha = 0.10$ level for the shorter time horizons.

Conclusions

Theoretical implications

This study introduced a new approach to forecasting overnight demand by leveraging social media information. While traditional methods such as the naïve approach and the ARIMA model struggled with forecasting (Gričar and Bojnec, 2019), boosting methods with no additional social media components could predict occupancy with greater precision. The XGBoost model's improvements on time horizons work counter by producing better results to forecast results Gunter and Önder (2016) showed on the FAVAR model's ability to predict

Test statistics horizon	Diebold-Mariano (DM) tests									
	H1: (1) < (*)				*H1*: (2) < (*)			*H1*: (3) < (*)		*H1*: (4) < (*)
	(2)	(3)	(4)	(5)	(3)	(4)	(5)	(4)	(5)	(5)
1 Day	**4.51**	**6.67**	**7.64**	**8.35**	**4.77**	**6.75**	**8.07**	1.08	*1.46*	0.51
4 Days	**1.88**	**4.22**	**6.03**	**7.09**	**3.27**	**5.25**	**6.55**	**1.93**	**2.98**	*1.49*
7 Days	*1.56*	**5.09**	**6.40**	**7.13**	**3.97**	**5.79**	**6.81**	**2.55**	**2.91**	*1.46*
14 Days	0.64	**2.77**	**4.10**	**5.44**	**2.41**	**4.15**	**5.45**	**2.21**	**3.51**	*1.57*

Notes: Test statistics in bold indicates significant at $\alpha < 0.05$, test statistics in *Italics* indicates significant at $\alpha < 0.10$. Models in the DM test are Naïve, ARIMA, XGBoost w/out Social, XGBoost w/Social and Segmented XGBoost w/Social. Test statistics are based on mean absolute deviation (MAD). Tests are applied on the out of time (OOT) data set

Table 2. Diebold-Mariano tests

shorter horizons. Furthermore, XGBoost can better predict both the shorter and longer horizons, which lends itself to a potential model that can augment precision while being well-generalized.

However, even a more robust boosting technique with inherited regularization controls methods such as XGBoost still struggled with potential overfitting issues on a dimensional data set (Chen and Guestrin, 2016). Furthermore, to address data struggles while using advanced learning algorithms, forecasters should follow our proposed framework or implement a set of extensive testing procedures. Forecasters need to take steps to reduce dimensionality without losing information. Our proposed framework further displayed that we could improve forecasts' precision and stability by understanding the underlying data and appropriate statistical methods.

Our model integrated social network data to improve forecasts and the hotel's managerial decisions on pricing. Many challenges transpire when scraping online social network data because of the required data cleaning; however, the inclusion of sentiment analysis provides a moderate improvement to demand to forecast. We incorporated sentiments by analyzed data from Twitter and SocialMention. It predicted overnight demand for a major hotel chain in the US hotel market, using a data set sample of 102,000 observations from the resort's PMS. Using such big data samples and incorporating many parameters creates challenges in running machine learning forecasting models. Hence, we introduced a hierarchical clustering method first to reduce the data, which allowed us to assess stability and robustness with each of the model forecasts.

Practical implications

Our study contributes to hospitality literature on a methodological approach based on its novel machine learning approach to demand forecasting while using social network information. This study also has managerial implications for hotel revenue managers. By following an empirical approach to use high-volume data, we can better apply an analysis of customer sentiments toward demand forecasting. In examining the trends, we found the longitudinal data to display a seasonality pattern, with spring as the peak season and fall as off-peak. The resort also exhibited more demand during the weekends. From a forecasting point, these patterns can improve capacity utilization and pricing recommendations.

By having said that, we first looked at forecasting demand without incorporating sentiment analysis data using the naïve approach, the results of which were not encouraging. The machine learning XGBoost model showed a 70% improvement over the ARIMA in the MAPE when used on a 1-day and a 30% improvement in the MAPE on a 14-day horizon. It indicates that advanced machine learning techniques outperform all other models. Following a sentiment analysis to the XGBoost model, for which the sentiment variables contributed 8.2%. This retuned XGBoost model to the OOT sample resulted in considerable improvements to MAPE, yet improvements to RMSE, compared to the XGBoost model. When we segmented the data into four clusters and forecasted using the XGBoost model with social media inputs, the differences in each segment's gains yielded noteworthy results. For instance, more recent daily pickups (i.e. within the previous 28 days) impacted the groups and OTA/GDS/phone segments.

In contrast, older daily pickups (between 90 and 180 days prior) contributed more to direct web bookings. The trend inputs' total contribution also differed by a customer segment. The social sentiment being most predictive with the direct website segment while least predictive with the walk-in's segment. Again, from a revenue manager's point-of-view, the result underlines the need for an adequate targeted customer market. Finally, the last noteworthy result reflected a minimal improvement in the forecasted MAPE when using the

segmented XGBoost on a 1-day horizon, compared to the same variables with the non-segmented XGBoost. The segmented XGBoost model helps correct the residuals in the forecast with limited precision. Another potential advantage is the ability to be more robust to market deviations that alter rooms' expected distribution. As non-segmented models' distributions are based on the entire population, drastic shifts can be made in the customer base or their resulting behavior such as the COVID-19 pandemic, non-segmented models struggle with the shifts. While that can also impact segmented models, these distributions' shifts can be isolated to a segmented forecasting approach, diminishing market deviations. The ability to be more robust to market deviations will allow hospitality organizations to segmented machine learning forecasts with longer lifecycles to minimize development time. On the other hand, while applying these inputs represents a novel method in hospitality, revenue managers should be aware of the data and model complexity and additional challenges.

Limitations and further research

This study introduced a new approach to demand forecasting, however, limitations in the sentiment analysis exist. Not every property has a Twitter account, making it even more challenging. Data sourcing is also an issue: this study is limited to a data set provided by a major chain hotel, while our method's applications on more and diverse properties might produce other results. Future studies should compare properties in various continents. Future studies might also use different dictionaries or a bag-of-words model, separate from the three major text-mining dictionaries to establish the word validity of tweets, to eliminate the extent of spelling mistakes and data noise (Schneider and Gupta, 2016). Despite these limitations, this study's XGBoost forecasting modeling can also be applied to others, including ones that analyze other types of properties, if researchers advance it.

Notes

1. An optimized distributed gradient boosting that implements machine learning algorithms.
2. Forecasting techniques are evaluated using four metrics: mean absolute deviation (MAD), means square error (MSE), root means squared error (RMSE) and mean absolute percentage error (MAPE) for forecast horizons h = 1, 4, 7, 14 days ahead. Based on a recursive forecasting approach, the prediction for the prior horizon is used as an input for predicting the next horizon.
3. The total sentiments in the ratio are higher than the number of cleansed tweets, as a tweet may contain both positive and negative sentiments.

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